Lab 01: Basic LAN Setup

2 PCs · 1 Router · 1 Switch

Lab #1	**Topic:** Networking Fundamentals	**Tool:** Cisco Packet Tracer	**Cert:** CompTIA Network+

1. Lab Objectives

- Build a simple Local Area Network (LAN) using two PCs, a switch, and a router.
- Assign static IP addresses to end devices and verify Layer 3 connectivity.
- Understand the role of each network device (PC, Switch, Router) in a LAN.
- Use the **ping** command to confirm end-to-end communication.
- Relate the lab to CompTIA Network+ objective: 2.2 – Compare and contrast networking devices.

2. Network Topology

The topology consists of two PCs connected to a single Layer 2 switch. The switch uplinks to a router interface (Fa0/0) which provides the default gateway for the LAN. All devices share the 192.168.1.0/24 subnet.

Device	Model	Interface	IP Address	Subnet Mask	Default GW
PC0	PC-PT	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	PC-PT	Fa0	192.168.1.11	255.255.255.0	192.168.1.1
SW1	2960	—	—	—	—
R1	1841	Fa0/0	192.168.1.1	255.255.255.0	—

3. Step-by-Step Configuration

3.1 PC IP Configuration

On each PC navigate to Desktop → IP Configuration and enter the static values:

PC0 – IP	192.168.1.10
PC0 – Mask	255.255.255.0
PC0 – GW	192.168.1.1
PC1 – IP	192.168.1.11

PC1 – Mask	255.255.255.0
PC1 – GW	192.168.1.1

3.2 Router R1 Configuration

Access R1 via CLI and configure the LAN interface:

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# exit
R1# write memory
```

4. Verification

From PC0 open the Command Prompt (Desktop → Command Prompt) and run:

```
C:\> ping 192.168.1.11   ! PC0 → PC1
C:\> ping 192.168.1.1    ! PC0 → Router
```

A successful ping returns: *Reply from ... bytes=32 time<1ms TTL=...* Four replies with 0% packet loss confirms full connectivity.

5. Key Takeaways

- Switches operate at Layer 2 and forward frames using MAC addresses; they do not need an IP.
- Routers operate at Layer 3 and require an IP address on each interface to route packets.
- A default gateway is the router IP that hosts send traffic to when the destination is outside the local subnet.
- Static IP addressing is reliable for small, fixed networks but does not scale well.
- Always use **no shutdown** to bring a router interface up — interfaces are down by default.